

Pixel Interface v3 - D1 STATUS LED Polarity Note

This note records the verified assembly orientation for the tiny STATUS LED on the pixel_interface_v3 PCB.

Mark the board: + by the round D1 pad, - by the square D1 pad.

D1 PCB pad	LED lead	Polarity	Net / function
Pin 2, round pad	Long lead	Anode (+)	STATUS / driven signal side
Pin 1, square pad	Short lead / flat side	Cathode (-)	R1 / resistor-to-GND side

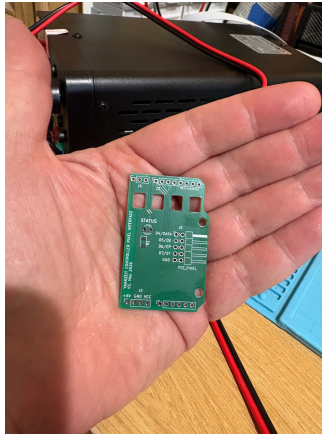
Verified current path:

STATUS pin -> D1 pin 2 / round pad / long lead / anode -> D1 pin 1 / square pad / short lead / cathode -> R1 -> GND.

Important lesson: ignore left/right from the schematic when assembling this board. The PCB footprint/pin mapping wins.

The KiCad DRC can pass this design because the nets are internally consistent. The issue is assembly polarity expectation: the square pad is D1 pin 1 and is the cathode on this PCB.

Reference photo - board is very small; use only a terse + / - mark near D1:



Suggested physical marking on each existing PCB: a tiny "+" at the round pad and a tiny "-" at the square pad.